AMAZON SALES & REVENUE ANALYSIS

Business Performance Analysis Report

Tools: Excel, Python, PostgreSQL, Power BI

Dataset: Amazon Sales Dataset

1. Executive Summary

This report evaluates Amazon sales and order data to assess business performance across transaction value, product demand, order outcomes, seller activity, payment methods, and geographic markets. The analysis was conducted using Excel and Python for data preparation, PostgreSQL for SQL-based analysis, and Power BI for performance visualisation.

The dataset comprises 100,000 order records, generating approximately 91.8 million in sales across 300,000 units sold. Average order value is approximately 918. Delivered orders account for 74,628 records, or 74.63% of all orders.

Electronics leads the analysed categories in both total quantity sold and reported sales value. Texas and California are the largest state-level contributors to sales, while seller-level analysis identifies SELL01099 as having the highest order count. Product-level analysis also identifies items with comparatively high combined cancellation and return rates.

The accompanying Power BI dashboard provides an overview of these measures through interactive KPIs, sales trends, category and brand comparisons, order-status distribution, and state-level sales visualisation. The findings establish a performance baseline and identify areas for further commercial and operational investigation.

2. Introduction

Transaction data can provide valuable insight into sales performance and operational outcomes when it is prepared, analysed, and presented systematically. This project applies a structured analytics workflow to Amazon order data to convert transaction records into business-relevant measures.

The report presents the analytical process, key findings, and dashboard design. It is intended to support performance monitoring and identify areas where further investigation may improve business understanding.

3. Project Objectives

The analysis aims to:

- Establish baseline sales and order performance.
- Identify leading categories and brands by sales and quantity.
- Examine order-status distribution and cancellation patterns.
- Compare geographic sales performance.
- Assess seller order volumes and payment-method performance.

- Identify products with elevated cancellation and return rates.
- Present key measures in an interactive Power BI dashboard.

4. Dataset and Methodology

4.1 Dataset Overview

The dataset contains 100,000 order records with fields covering orders, customers, products, transaction values, payment methods, order status, geography, and sellers.

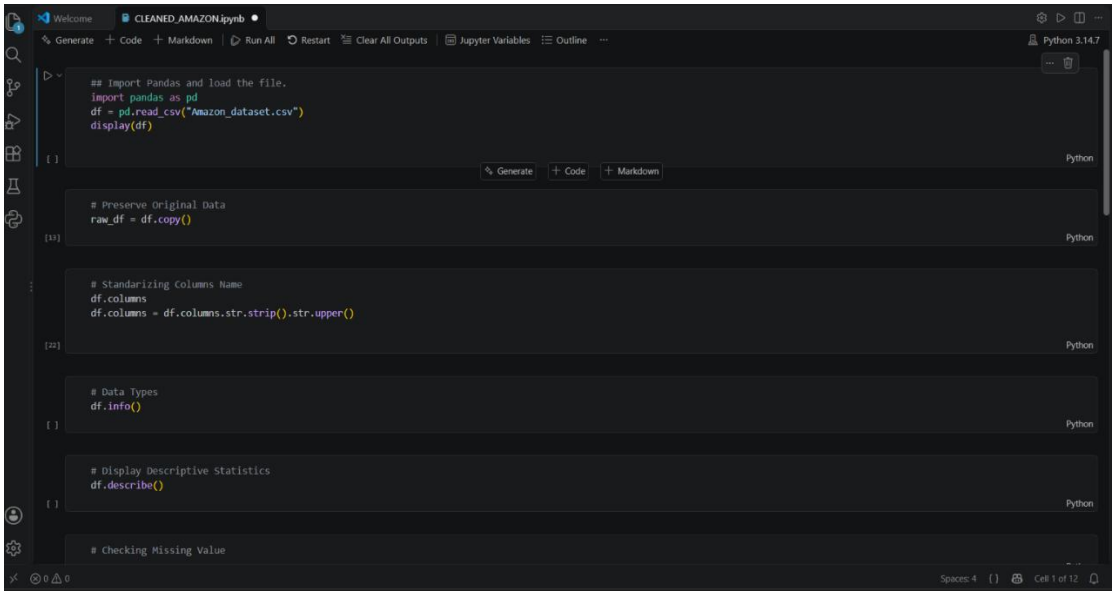
Data Area	**Key Fields**
Orders	OrderID, OrderDate, OrderStatus
Customers	CustomerID, CustomerName
Products	ProductID, ProductName, Category, Brand
Transaction values	Quantity, UnitPrice, Discount, Tax, ShippingCost, TotalAmount
Payment	PaymentMethod
Geography	City, State, Country
Sellers	SellerID

4.2 Data Preparation

Excel and Python were used to inspect and prepare the dataset. The preparation process included reviewing column structure and data types, standardising column names, trimming whitespace, identifying inconsistent missing-value representations, and checking for duplicate records.

The prepared dataset was loaded into PostgreSQL for structured analysis. These steps established a consistent basis for aggregating and comparing business measures.

Figure 1. Python-based data preparation and cleaning workflow

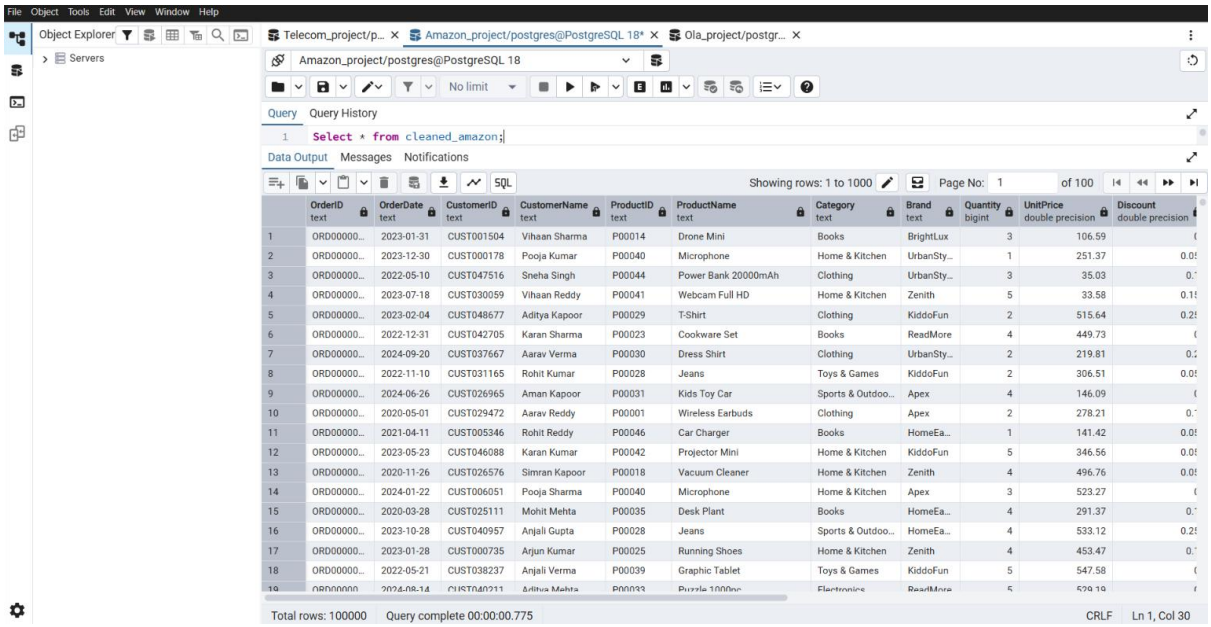


4.3 Analytical Method

SQL queries and views were used to analyse sales and quantity by category and brand, order-status distribution, city-level performance, seller order counts, cancellation rates, state-level sales contribution, payment-method performance, product cancellation and return rates, and brand discount patterns.

The resulting measures were selected for inclusion in the Power BI dashboard.

Figure 2. SQL-based analysis of Amazon sales data in PostgreSQL



5. Business Performance Analysis

5.1 Overall Performance

Key Performance Indicator	Result
Total Sales	91.8M
Total Orders	100,000
Total Quantity Sold	300,000
Average Order Value	918

These indicators provide the baseline for monitoring transaction value, order volume, and units sold. Average order value complements total sales by showing the average transaction value across recorded orders.

5.2 Category and Brand Performance

Electronics recorded the highest total quantity sold, at 50,696 units. It also recorded the highest reported category sales, at 15,584,217.18.

At brand level, CoreTech recorded the highest total quantity sold, at 30,430 units. Within the above-average category-brand combinations, Toys & Games–CoreTech had the highest quantity, at 5,298 units.

These results identify leading category and brand combinations by the measures analysed. They do not, by themselves, establish profitability or explain the drivers of demand.

5.3 Order Status and Fulfilment

Order Status	Orders	Share of Orders
Delivered	74,628	74.63%
Shipped	15,192	15.19%
Pending	4,103	4.10%
Returned	3,049	3.05%
Cancelled	3,028	3.03%

Delivered orders represent the largest share of recorded order outcomes. Shipped, pending, returned, and cancelled orders provide additional operational categories for monitoring.

5.4 Geographic Performance

Texas contributed 22,862,540.54 in sales, representing 24.90% of total sales. California contributed 18,231,196.79, or 19.85%.

Among cities with average order values above the overall average, Denver recorded the highest average order value at 929.37, followed closely by Charlotte at 929.10.

Charlotte recorded the highest number of delivered orders among the cities examined, with 3,845.

5.5 Seller Performance

Seller SELL01099 recorded the highest order count, with 78 orders.

The state-level seller-count analysis identified California and Texas as jointly having the highest number of distinct sellers, at 1,999 each. This measure reflects seller count by state, not seller sales ranking.

5.6 Payment Method Performance

Debit Card and UPI were the payment methods identified as having above-average sales per order.

Payment Method	Orders	Total Sales	Average Sales per Order
Debit Card	20,024	18,538,678.53	925.82
UPI	15,066	13,896,028.55	922.34

Debit Card recorded the higher average sales per order among these methods.

5.7 Cancellation and Return Patterns

Electronics recorded the highest cancellation rate among categories exceeding the overall cancellation rate, at 3.23%. Toys & Games recorded 3.10%, and Home & Kitchen recorded 3.06%.

At product level, Dress Shirt had the highest combined cancellation and return rate among the ten products returned by the analysis, at 7.32%, affecting 147 orders. Yoga Mat followed at 7.20%, also affecting 147 orders.

These results identify areas for further review; the analysis does not establish the causes of cancellations or returns.

5.8 Discount Patterns

FitLife recorded the highest average discount among the brands above the overall average, at 7.58%. UrbanStyle followed at 7.52%.

Discount levels should be evaluated alongside sales volume, revenue, and margin data before drawing conclusions about commercial effectiveness.

6. Power BI Dashboard and Performance Monitoring

The Power BI dashboard consolidates the principal findings into an interactive view of sales performance.

6.1 Key Performance Indicators

The dashboard displays four headline KPIs:

- Total Sales
- Total Orders
- Total Quantity Sold
- Average Order Value

These indicators provide a concise overview of transaction value, volume, and average order size.

Figure 3. Amazon Sales & Revenue Analysis dashboard



6.2 Interactive Filters

The dashboard includes slicers for Category, Brand, State, Payment Method, and Order Status. These filters allow users to examine performance across selected business dimensions.

6.3 Visualisations

Visualisation	Business Purpose
Stacked area chart — Sales Trend Over Time	Examine sales patterns across the reporting period
Pie chart — Sales by Category	Compare category contributions to sales
Clustered bar chart — Top 5 Brands by Sales	Compare the five leading brands by sales value
Pie chart — Order Status Distribution	Review the composition of order outcomes
Filled map — Sales by State	Explore geographic sales distribution

Together, these visuals provide a high-level performance overview and support focused exploration of the underlying sales data.

7. Key Findings and Management Considerations

The analysis highlights the following areas for management attention:

1. **Category performance:** Electronics leads in quantity sold and reported category sales. Category performance should be monitored alongside margin and cost measures.
2. **Geographic concentration:** Texas and California account for the largest reported state-level sales contributions. Their performance merits continued monitoring.
3. **Order outcomes:** Delivered orders represent 74.63% of records. The remaining order statuses should be reviewed to understand operational progression and exceptions.
4. **Cancellation and return monitoring:** Electronics has the highest cancellation rate among the categories identified as above the overall rate. Dress Shirt has the highest combined product-level cancellation and return rate among the reported top ten.

5. **Seller and payment performance:** SELL01099 leads by order count, while Debit Card has the highest average sales per order among the above-average payment methods.
6. **Discount evaluation:** FitLife has the highest average discount among the brands exceeding the overall average. Discount decisions should be assessed against profitability measures.

These considerations are intended to guide further investigation and performance monitoring rather than assert causal explanations unsupported by the dataset.

8. Analytical Limitations

- The analysis is based on the fields and records available in the supplied dataset.
- Sales values are reported in the dataset's recorded units; no currency conversion has been applied.
- The analysis does not include product costs, operating expenses, or profit margins. Sales value should not be interpreted as profit.
- Cancellation and return rates are descriptive and do not identify their underlying causes.
- Seller performance is assessed using order counts and seller counts; a complete seller profitability or service-quality assessment would require additional measures.
- Findings reflect the analysed dataset and reporting period and should not be generalised beyond its scope without further validation.

9. Conclusion

This project transformed Amazon order data into a structured view of sales, product, geographic, seller, payment, and order-status performance. The analysis established baseline KPIs, identified leading categories and brands, highlighted major state-level sales contributors, and surfaced cancellation and return patterns for further review.

The Power BI dashboard provides an accessible means of monitoring these measures and exploring performance through interactive filters. Together, the SQL analysis and dashboard offer a practical foundation for data-informed business review, while further profitability and operational data would be required to support deeper causal or financial conclusions.